

Organized by:



FAKULTAS TEKNIK
Universitas Malikussaleh

SENIOR HIGH SCHOOL EDITION GUIDEBOOK

IPDCEC 2026

INTERNATIONAL POSTER
SCIENTIFIC POSTER COMPETITION
FOR SENIOR HIGH SCHOOL LEVEL 2026

*"Industry-Based Chemical Plant Design
for a Sustainable Global Future"*



IPDCEC

INTERNATIONAL PLANT DESIGN
CHEMICAL ENGINEERING COMPETITION



www.ipdcec.online



[@ipdcec2026](https://www.instagram.com/ipdcec2026)



ipdcec2026@gmail.com

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(ENGLISH VERSION) IPDCEC GUIDEBOOK

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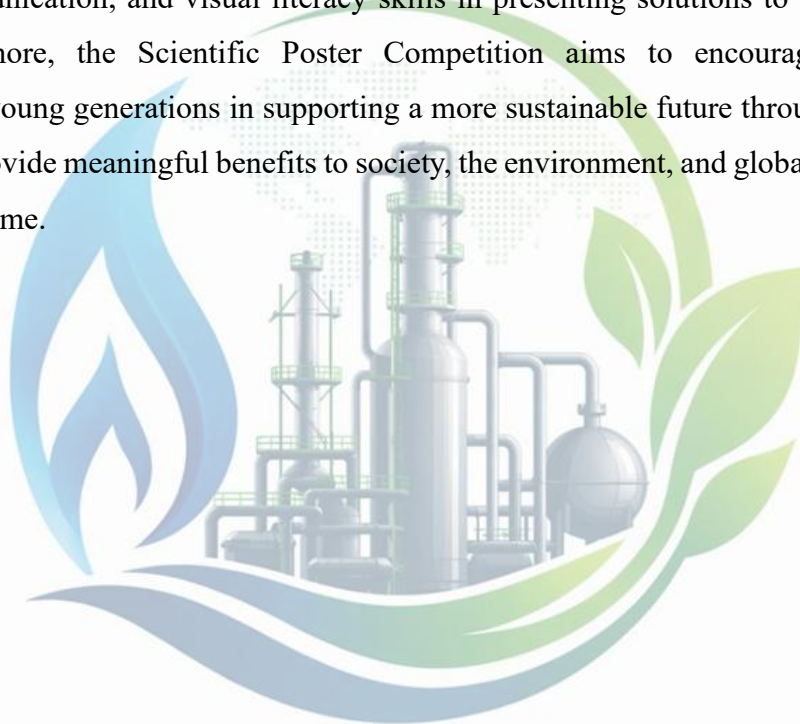


ipdcec2026@gmail.com

GENERAL DESCRIPTION

Scientific Poster Competition is an international-level competition organized as part of the International Plant Design Chemical Engineering Competition (IPDCEC) 2026 by the Department of Chemical Engineering, Faculty of Engineering, Universitas Malikussaleh. This competition serves as a platform for high school students (Senior High School, Vocational High School, Islamic Senior High School, or equivalent) from various countries to express their ideas, creativity, and innovations in addressing global challenges related to the environment, energy, technology, and sustainable development through scientific posters that are informative, engaging, and easily understood by the general public.

Through this competition, participants are expected to enhance their critical thinking, scientific communication, and visual literacy skills in presenting solutions to various global issues. Furthermore, the Scientific Poster Competition aims to encourage the active involvement of young generations in supporting a more sustainable future through innovative ideas that can provide meaningful benefits to society, the environment, and global development in the years to come.



BACKGROUND

Today, the world is facing increasingly complex global challenges, including climate change, environmental pollution, energy crises, and the growing volume of waste generated by human activities. According to the United Nations Environment Programme (UNEP), global municipal waste generation is projected to increase from 2.1 billion tonnes in 2023 to 3.8 billion tonnes by 2050 if more effective and sustainable waste management efforts are not implemented (UNEP, 2024). Furthermore, these environmental issues contribute to the ongoing planetary crisis, which encompasses climate change, biodiversity loss, and increasing pollution and waste, all of which threaten human well-being and the sustainability of ecosystems (UNEP, 2024).

Amid these challenges, young people play a crucial role as agents of change in developing innovative and sustainable solutions. UNESCO emphasizes that youth engagement in generating ideas, fostering innovation, and taking concrete action is essential for advancing sustainable development and addressing future global environmental challenges (UNESCO, 2024). Therefore, the Scientific Poster Competition is organized as a platform for high school students (Senior High School, Vocational High School, Islamic Senior High School, or equivalent) from around the world to express their ideas, creativity, and innovative solutions through scientific poster presentations. Through this competition, participants are expected to raise public awareness of global issues while inspiring the development of ideas that contribute to a greener, more sustainable, and environmentally friendly future.



THEME

"Industrial-Based Chemical Plant Design for a Sustainable Global Future"

SUBTHEMES

1. Renewable Energy
2. Waste Management
3. Green Technology
4. Sustainable Industry
5. Climate Change and Its Solutions
6. Circular Economy



TERMS AND CONDITIONS

1. The competition is open to active high school/vocational high school students (Senior High School, Vocational High School, Islamic Senior High School, or equivalent) from schools worldwide.
2. The competition may be entered individually or as a team consisting of a maximum of three (3) participants from the same school.
3. Each participant may only be a member of one (1) team.
4. Each team is allowed to submit only one (1) poster entry.
5. Submitted works must be original and must not have previously won any other competition.
6. Posters must be relevant to the theme and subtheme specified by the organizing committee.
7. Participants may choose to use Artificial Intelligence (AI) tools in the creation of their work or create their work without AI assistance.
8. Entries must not contain content related to discrimination, ethnicity, religion, race, pornography, violence, hate speech, or any form of copyright infringement.
9. The submitted work must be published through the participant's Instagram account in accordance with the publication requirements established by the organizing committee.
10. Participants Instagram accounts must remain public throughout the evaluation period.
11. The decisions of the judging panel are final and binding and cannot be appealed.



WORK SUBMISSION REQUIREMENTS

1. The poster must be created in a digital format.
2. Poster size: A3 (297 mm × 420 mm).
3. The poster must be in portrait orientation.
4. Accepted file formats: PDF or JPG/JPEG, (Max. 10 MB)
5. The file must be uploaded using the following naming format:
(Team Name_School Name_Poster_IPDCEC 2026)
6. Minimum resolution: 300 dpi.
7. The poster may be prepared in either Indonesian or English.
8. The poster must include:
 - Title
 - The IPDCEC logo and the participant's school logo, which must be placed in the upper-right corner of the poster
 - The issue or problem being addressed
 - The proposed solution or innovation
 - The expected impact or benefits
 - References (if applicable)
 - If Artificial Intelligence (AI) tools are used, participants must clearly state the type of AI used in the work description section (e.g., ChatGPT, Gemini, Claude, Midjourney, DALL·E, Canva AI, Adobe Firefly, etc.).



INSTAGRAM PUBLICATION REQUIREMENTS

1. The poster must be uploaded to the participant's Instagram feed.
2. The caption must include:
 - The title of the work
 - Participant's/team's name
 - School name
 - Official competition hashtags
3. The official organizer's Instagram account must be tagged in the post.
4. The participant's Instagram account must remain public until the winners are officially announced.



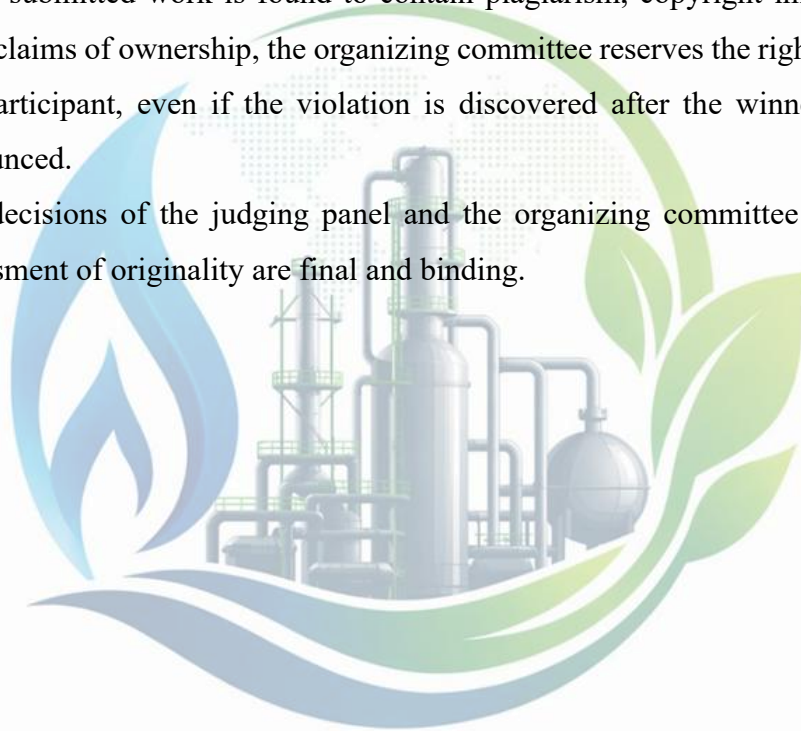
REGISTRATION REQUIREMENTS

1. Participants are required to follow the official Instagram account @ipdcec2026, upload the official competition twibbon, and share the competition promotional poster while tagging @ipdcec2026.
2. Registration and submission of entries must be completed through the registration form provided by the organizing committee.
3. Participants are required to upload the following documents:
 - Student Identification Card or an Official Certificate of Enrollment from their school;
 - Proof of following the Instagram account @ipdcec2026;
 - Proof of twibbon upload;
 - Proof of sharing the competition promotional poster;
 - Poster submission in accordance with the specified competition requirements.
4. All uploaded documents must comply with the specified format and be clearly legible.
5. Submitted data cannot be modified after the registration and submission deadline.
6. Participants are fully responsible for the accuracy of the information provided during the registration process.
7. Participants are required to regularly monitor announcements and updates through IPDCEC's official social media platforms.
8. The registration fee is **USD 1 (one United States Dollar)** or the equivalent amount in local currency.
9. Payment of the registration fee is made via:
 - **Bank Transfer** for participants from Indonesia.
BSI: 7242691509 (Syaidatul Amira)
10. Registration and entry submission open on **July 9, 2026** and close on **July 22, 2026 at 23:59 WIB** (Western Indonesia Time).
11. For further information, participants may contact:
 - Diah Sandi Ayu Sinaga : (+62 831-6759-4356)
 - Wirda Zhafira Lubis : (+62 813-6202-3850)



WORK ORIGINALITY REQUIREMENTS

1. The submitted work must be the participant's original creation and must not have previously won any similar competition.
2. Participants are permitted to use design software and/or Artificial Intelligence (AI) as supporting tools during the creation process.
3. The use of AI does not exempt participants from the responsibility of producing an original work with a self-developed concept that does not infringe upon the intellectual property rights of others.
4. Participants are prohibited from using, copying, or modifying another person's work without proper authorization.
5. If the submitted work is found to contain plagiarism, copyright infringement, or false claims of ownership, the organizing committee reserves the right to disqualify the participant, even if the violation is discovered after the winners have been announced.
6. The decisions of the judging panel and the organizing committee regarding the assessment of originality are final and binding.



TIMELINE

Activity	Date
Registration and Submission Opening	July 10, 2026
Registration and Submission Deadline	July 22, 2026
Judging Period	July 23–28, 2026
Winners Announcement	August 1, 2026, at Universitas Malikussaleh



ASSESSMENT CRITERIA

No.	Evaluation Criteria	Evaluation Indicators	Weight
1.	Theme Relevance	• The poster is relevant to the main theme and one of the competition subthemes. • The proposed idea is relevant to the scientific issue or problem being addressed. • The topic demonstrates urgency and relevance to real-world issues in society, the environment, technology, or industry.	25%
2.	Creativity and Innovation	• The proposed idea or solution demonstrates originality and innovation. • The poster goes beyond common ideas by presenting a unique perspective or innovative approach. • The proposed solution is engaging, practical, and has the potential for further development.	25%
3.	Content Quality	• The information presented is accurate, logical, and supported by scientific data or evidence. • The poster presents a clear flow of discussion, including the background, objectives, methods or proposed idea, results or solution, and conclusion. • Appropriate theories, data, or references are used to support the content of the poster.	25%
4.	Poster Visual Design	• The poster layout is neat, visually appealing, and easy to read. • Appropriate use of colors, fonts, images, illustrations, graphs, and icons that align with the theme. • A balanced composition of text and visuals that enhances readability without being overcrowded. • The poster has a clear and professional visual identity.	15%
5.	Impact and Practicality	• The feasibility of the proposed idea for future implementation. • The potential positive impact of the solution on the environment, industry, or society.	10%
	Total Score		100%



CONTACT PERSON

For further information, please contact:

Syaidatul Amira	: (+62 812-6705-5711)
Diah Sandi Ayu Sinaga	: (+62 831-6759-4356)
Wirda Zhafira Lubis	: (+62 813-6202-3850)
Email	: ipdcec2026@gmail.com
Instagram	: @ipdcec2026
Tiktok	: @ipdcec2026
Youtube	: @ipdcec2026
LinkedIn	: @International Plant Design Chemical Engineering Competition (IPDCEC)



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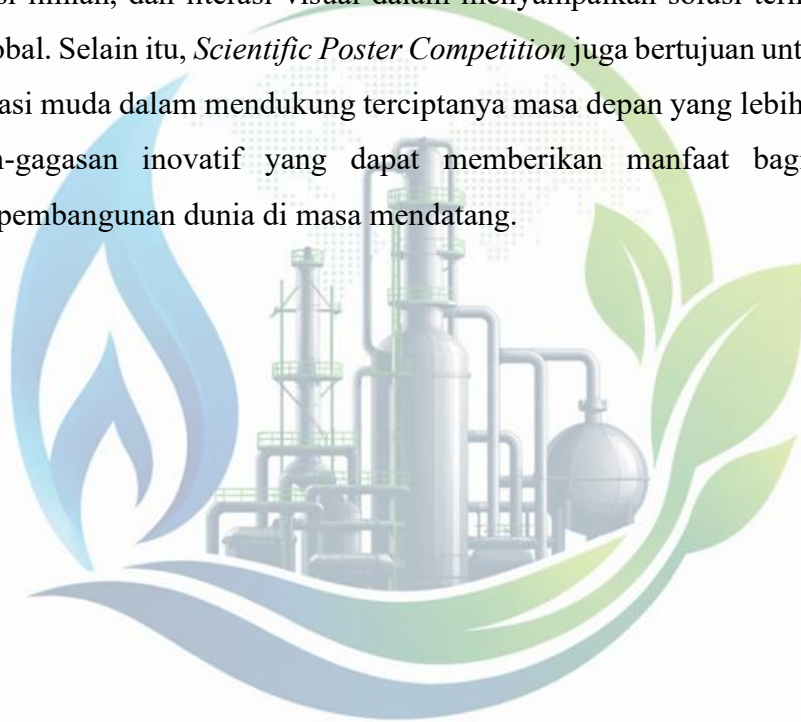


ipdcec2026@gmail.com

DESKRIPSI UMUM

Scientific Poster Competition merupakan kompetisi tingkat internasional yang diselenggarakan sebagai bagian dari rangkaian kegiatan *International Plant Design Chemical Engineering Competition* (IPDCEC) 2026 oleh Jurusan Teknik Kimia, Fakultas Teknik, Universitas Malikussaleh. Kompetisi ini menjadi wadah bagi siswa SMA/SMK/MA sederajat dari berbagai negara untuk menuangkan ide, kreativitas, serta inovasi dalam menyikapi berbagai tantangan global yang berkaitan dengan lingkungan, energi, teknologi, dan pembangunan berkelanjutan melalui media poster ilmiah yang informatif, menarik, dan mudah dipahami oleh masyarakat luas.

Melalui kompetisi ini, peserta diharapkan mampu meningkatkan kemampuan berpikir kritis, komunikasi ilmiah, dan literasi visual dalam menyampaikan solusi terhadap berbagai permasalahan global. Selain itu, *Scientific Poster Competition* juga bertujuan untuk mendorong peran aktif generasi muda dalam mendukung terciptanya masa depan yang lebih berkelanjutan melalui gagasan-gagasan inovatif yang dapat memberikan manfaat bagi masyarakat, lingkungan, dan pembangunan dunia di masa mendatang.



LATAR BELAKANG

Dunia saat ini menghadapi berbagai tantangan global yang semakin kompleks, seperti perubahan iklim, pencemaran lingkungan, krisis energi, serta peningkatan jumlah limbah yang dihasilkan dari aktivitas manusia. Menurut *United Nations Environment Programme* (UNEP), jumlah sampah perkotaan global diperkirakan akan meningkat dari 2,1 miliar ton pada tahun 2023 menjadi 3,8 miliar ton pada tahun 2050 apabila tidak dilakukan upaya pengelolaan yang lebih efektif dan berkelanjutan (UNEP, 2024). Selain itu, berbagai permasalahan lingkungan tersebut turut berkontribusi terhadap krisis planet yang mencakup perubahan iklim, hilangnya keanekaragaman hayati, serta meningkatnya polusi dan limbah yang berdampak pada kualitas hidup manusia dan keberlanjutan ekosistem (UNEP, 2024).

Di tengah berbagai tantangan tersebut, generasi muda memiliki peran penting sebagai agen perubahan dalam menciptakan solusi yang inovatif dan berkelanjutan. UNESCO menegaskan bahwa keterlibatan generasi muda dalam pengembangan ide, inovasi, dan aksi nyata sangat diperlukan untuk mendukung pembangunan berkelanjutan dan menghadapi tantangan lingkungan global di masa depan (UNESCO, 2024). Oleh karena itu, *Scientific Poster Competition* diselenggarakan sebagai wadah bagi siswa SMA/SMK/MA sederajat dari seluruh dunia untuk menuangkan gagasan, kreativitas, serta solusi inovatif melalui media poster ilmiah. Melalui kompetisi ini, peserta diharapkan mampu meningkatkan kesadaran masyarakat terhadap berbagai isu global sekaligus mendorong lahirnya ide-ide yang dapat berkontribusi dalam mewujudkan masa depan yang lebih hijau, berkelanjutan, dan ramah lingkungan.



TEMA

” Perancangan Pabrik Kimia Berbasis Industri untuk Masa Depan Global yang Berkelanjutan”

SUBTEMA

1. Energi Terbarukan
2. Pengelolaan Limbah
3. Teknologi Hijau
4. Industri Berkelanjutan
5. Perubahan Iklim dan Solusinya
6. Ekonomi Sirkular



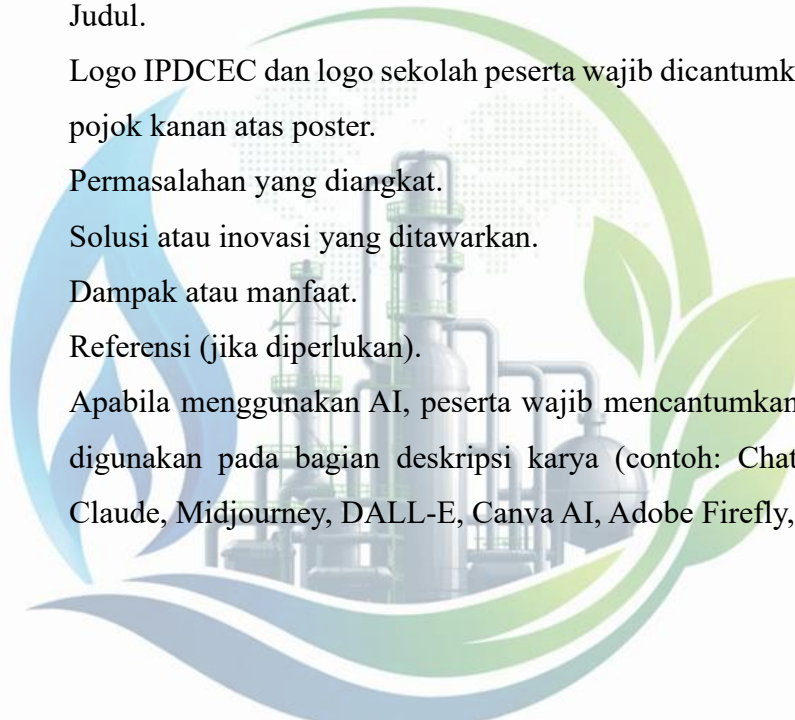
SYARAT & KETENTUAN

1. Kompetisi terbuka bagi siswa aktif SMA/SMK/MA atau sederajat dari seluruh sekolah di dunia.
2. Kompetisi dapat diikuti secara individu maupun tim yang terdiri dari maksimal 3 (tiga) orang peserta dari sekolah yang sama.
3. Setiap peserta hanya diperbolehkan tergabung dalam 1 (satu) tim.
4. Setiap tim hanya diperbolehkan mengirimkan 1 (satu) karya poster.
5. Karya yang dikirimkan harus merupakan karya orisinal dan belum pernah menjadi juara pada kompetisi lain.
6. Poster harus sesuai dengan tema dan subtema yang telah ditentukan panitia.
7. Peserta diperbolehkan menggunakan Artificial Intelligence (AI) dalam proses pembuatan karya maupun tidak menggunakannya.
8. Karya tidak boleh mengandung unsur SARA, pornografi, kekerasan, ujaran kebencian, maupun pelanggaran hak cipta.
9. Karya wajib dipublikasikan melalui akun Instagram peserta sesuai ketentuan yang ditetapkan panitia.
10. Akun Instagram peserta tidak boleh dalam kondisi private selama masa penilaian.
11. Keputusan dewan juri bersifat final dan tidak dapat diganggu gugat.



KETENTUAN KARYA

1. Poster dibuat dalam format digital.
2. Ukuran poster: A3 (297 mm × 420 mm).
3. Orientasi portrait.
4. Format file: PDF atau JPG/JPEG, (Max. 10 MB)
5. File di unggah dengan format:
(Nama Tim_Asal Sekolah_Poster_IPDCEC 2026)
6. Resolusi minimum: 300 dpi.
7. Bahasa yang digunakan dapat berupa Bahasa Indonesia atau Bahasa Inggris.
8. Poster harus memuat:
 - Judul.
 - Logo IPDCEC dan logo sekolah peserta wajib dicantumkan pada bagian pojok kanan atas poster.
 - Permasalahan yang diangkat.
 - Solusi atau inovasi yang ditawarkan.
 - Dampak atau manfaat.
 - Referensi (jika diperlukan).
 - Apabila menggunakan AI, peserta wajib mencantumkan jenis AI yang digunakan pada bagian deskripsi karya (contoh: ChatGPT, Gemini, Claude, Midjourney, DALL-E, Canva AI, Adobe Firefly, dan lain-lain).



KETENTUAN ORISINALITAS KARYA

1. Karya yang dilombakan harus merupakan hasil kreativitas peserta dan belum pernah menjadi juara pada kompetisi sejenis.
2. Peserta diperbolehkan menggunakan perangkat lunak desain maupun Artificial Intelligence (AI) sebagai alat bantu dalam proses pembuatan karya.
3. Penggunaan AI tidak menghilangkan kewajiban peserta untuk menghasilkan karya yang orisinal, memiliki konsep yang dikembangkan sendiri, serta tidak melanggar hak cipta pihak lain.
4. Dilarang menggunakan, menyalin, atau memodifikasi karya milik pihak lain tanpa izin yang sah.
5. Apabila terbukti terdapat unsur plagiarisme, pelanggaran hak cipta, atau pemalsuan kepemilikan karya, panitia berhak mendiskualifikasi peserta meskipun pelanggaran tersebut ditemukan setelah pengumuman pemenang.
6. Keputusan dewan juri dan panitia terkait penilaian orisinalitas bersifat final.



KETENTUAN PUBLIKASI INSTAGRAM

1. Poster diunggah pada feed Instagram peserta.
2. Caption harus memuat:
 - Judul karya
 - Nama peserta/tim
 - Asal sekolah
 - Hashtag resmi kompetisi
3. Menandai akun resmi penyelenggara.
4. Akun Instagram harus tetap publik hingga pengumuman pemenang.



KETENTUAN REGISTRASI

1. Peserta diwajibkan untuk mengikuti akun Instagram resmi @ipdcec2026, mengunggah twibbon resmi kompetisi, serta membagikan poster publikasi kompetisi dengan menandai akun @ipdcec2026.
2. Pendaftaran dan pengumpulan karya dilakukan melalui formulir pendaftaran yang telah disediakan oleh panitia.
3. Peserta wajib mengunggah dokumen yang dipersyaratkan, meliputi:
 - Kartu Pelajar atau Surat Keterangan Aktif dari sekolah;
 - Bukti mengikuti akun Instagram @ipdcec2026;
 - Bukti unggah twibbon;
 - Bukti unggah poster publikasi kompetisi;
 - Karya poster sesuai dengan ketentuan yang telah ditetapkan.
4. Seluruh dokumen yang diunggah harus sesuai dengan format yang telah ditentukan serta dapat dibaca dengan jelas.
5. Data yang telah dikirimkan tidak dapat diubah setelah batas akhir pendaftaran dan pengumpulan karya.
6. Peserta bertanggung jawab penuh atas keakuratan data yang diberikan selama proses pendaftaran.
7. Peserta diwajibkan untuk memantau informasi dan pengumuman melalui media sosial resmi IPDCEC secara berkala.
8. Biaya pendaftaran sebesar USD 1 (satu dolar Amerika Serikat) atau setara dalam mata uang lokal.
9. Pembayaran biaya pendaftaran dilakukan melalui:
10. Pembayaran biaya pendaftaran dilakukan melalui:
 - **Transfer Bank** untuk peserta dari Indonesia.
BSI: 7242691509 (Syaidatul Amira)
11. Pendaftaran dan pengumpulan karya dibuka pada 10 Juli 2026 dan ditutup pada 22 Juli 2026 pukul 23.59 WIB
12. Untuk informasi lebih lanjut, peserta dapat menghubungi:
Diah Sandi Ayu Sinaga : (+62 831-6759-4356)



Wirda Zhafira Lubis : (+62 813-6202-3850)



TIMELINE

Tahapan Kegiatan	Tanggal Pelaksanaan
Pembukaan Pendaftaran dan Pengumpulan Karya	10 Juli 2026
Penutupan Pendaftaran dan Pengumpulan	22 Juli 2026
Masa Penilaian	23–28 Juli 2026
Pengumuman Pemenang	01 Agustus 2026 di Universitas Malikussaleh



KRITERIA PENILAIAN

No	Kriteria Penilaian	Indikator Penilaian	Bobot
1.	Kesesuaian Tema	• Poster relevan dengan tema utama dan subtema lomba. • Gagasan yang diangkat sesuai dengan isu/permasalahan ilmiah yang dibahas. • Topik memiliki urgensi dan keterkaitan dengan kondisi nyata di Masyarakat, lingkungan, teknologi, atau industry.	25%
2.	Kreativitas dan Inovasi	• Ide atau solusi yang ditawarkan memiliki unsur kebaruan dan keunikan. • Poster tidak hanya mengulang gagasan umum, tetapi memberikan sudut pandang atau pendekatan baru. • Solusi yang ditawarkan menarik, aplikatif, dan berpotensi dikembangkan.	25%
3.	Kualitas Konten	• Informasi yang disampaikan akurat, logis, dan berbasis data/fakta ilmiah. • Terdapat alur pembahasan yang jelas, mulai dari latar belakang masalah, tujuan, metode/gagasan, hasil/solusi, hingga Kesimpulan. • Penggunaan teori, data, atau referensi mendukung poster.	25%
4.	Desain Visual Video	• Tata letak poster rapi, menarik, dan mudah dibaca. • Pemilihan warna, font, gambar, ilustrasi, grafik, dan ikon sesuai dengan tema. • Komposisi teks dan visual seimbang, tidak terlalu penuh, serta mendukung pemahaman isi. • Poster memiliki identitas visual yang jelas dan profesional.	15%
5.	Dampak dan Kebermanfaatan	• Rasionalitas atau kelayakan (<i>feasibility</i>) ide untuk direalisasikan di masa depan. • Skala dampak positif solusi terhadap lingkungan, industri, atau masyarakat.	10%
		Total Penilaian	100%



CONTACT PERSON

Untuk informasi lebih lanjut, hubungi:

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Wirda Zhafira Lubis	: (+62 813-6202-3850)
Email	: ipdcec2026@gmail.com
Instagram	: @ipdcec2026
Tiktok	: @ipdcec2026
Youtube	: @ipdcec2026
LinkedIn	: @International Plant Design Chemical Engineering Competition (IPDCEC)

